

The scheme relies on support from the Memory Manager to retrieve page usage histories. It also calls on the Memory Manager to preload data and code from files on disk or from a paging file into the Standby List and assign priorities to pages. Thus it might pre-fetch code from Adobe Photoshop, such that when you launch Photoshop, it is started much faster than before.

When an application exits and frees up memory, SuperFetch asks the Memory Manager to fetch the applications associated data and code which is going to be flushed from the memory proper. This data is then stored by SuperFetch, such that when you need to resume a program, SuperFetch can bring all or most of it back into memory. Going back to our scenario with the anti-virus: SuperFetch would store the data and code of the Excel sheet when the anti-virus application is launched. It would also make sure that most of that code and data is back into memory before you return.

SuperFetch extends this support to other aspects of the system as well, aspects such as Hibernation and Fast User Switching. For hibernation, SuperFetch stores data and code in the hibernation file that it expects (based on historical records) will be accessed during the subsequent resume. In contrast, when you resume Windows XP, previously cached data must be reread from the disk when it is referenced.

3.8 Reliability, Security, And Power Management

3.8.1 ReadyBoost

On a typical system, I/O operations take a significant chunk of overall system processing. A hard drive read or write is almost as frequent as CPU to memory transfers. This highlights one of the biggest bottlenecks of our computer systems—hard disk access times. When compared to the speed at which the rest of the system operates—the processor, the memory system (RAM), the video